

When Inverse Data Outperforms: Exploring the Pitfalls of Mixed Data in Multi-Stage Fine-Tuning

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Abstract

Existing work has shown that o1-level performance can be achieved with limited data distillation, but most existing methods focus on unidirectional supervised fine-tuning (SFT), overlooking the intricate interplay between diverse reasoning patterns. In this paper, we construct **r1k**, a high-quality reverse reasoning dataset derived by inverting 1,000 forward examples from **s1k** (Muennighoff et al., 2025), and examine how SFT and Direct Preference Optimization (DPO) affect alignment under bidirectional reasoning objectives. SFT on **r1k** yields a **1.6%–6.8%** accuracy improvement over **s1k** across evaluated benchmarks. However, naively mixing forward and reverse data during SFT weakens the directional distinction. Although DPO can partially recover this distinction, it also suppresses less preferred reasoning paths by shifting the probability mass toward irrelevant outputs. These findings suggest that mixed reasoning data introduce conflicting supervision signals, underscoring the need for robust and direction-aware alignment strategies. Our code and data are available at: <https://github.com/16demi/ReasonAlign-analysis>.

1 Introduction

Recent studies show that Large Language Models (LLMs) can achieve strong reasoning performance by distilling knowledge from a small set of high-quality examples (Li et al., 2025). Methods like **s1** (Muennighoff et al., 2025) and **LIMO** (Ye et al., 2025) demonstrate that with just 817 to 1,000 curated samples, a 32B model can match or surpass larger systems. However, these approaches focus mainly on single-direction reasoning—solving problems step by step from question to answer. As shown in Figure 1, a model may learn to compute

the kinetic energy of a gas molecule from its temperature, but not the reverse: inferring temperature from energy.

This narrow focus overlooks a core aspect of human cognition: **the inherently bidirectional nature** of reasoning. Humans commonly engage in backward reasoning, particularly in goal-directed problem solving (Newell et al., 1972; Hawes et al., 2012). Rather than reasoning solely from premises to conclusions, people often begin with a desired outcome and work backward through intermediate steps to reach known facts (Senn and Sacramento, 2015). Motivated by this cognitive insight, recent studies have begun to explore reverse or backward reasoning in LLM. MathGenie (Lu et al., 2024) utilizes reverse derivation paths to improve robustness on math word problems. Iterative Question Composing (Liu et al., 2024) constructs intermediate subquestions that align with goal-driven, backward-style planning. In optimization modeling, OptiBench (Yang et al., 2025) promotes reflective and Socratic-style reformulations that partially embody reverse reasoning principles. While promising, these approaches remain constrained to short-context reasoning or domain-specific tasks. This leaves open the broader question of whether backward supervision can enhance long chain-of-thought (Wei et al., 2022) reasoning and generalize across diverse scenarios.

To investigate this, we construct a high-quality dataset, **r1k**, by systematically inverting 1,000 forward reasoning examples from **s1k** (Muennighoff et al., 2025). Reverse questions and reasoning paths are generated with cost-efficient DeepSeek-R1 model (Guo et al., 2025), without the need for expensive data collection, cleaning, or selection procedures. Fine-tuning on **r1k** yields an approximate 1.6%–6.8% improvement over **s1k**.

To further study the interaction between mixed data, we conducted extensive experiments on their combined effects. We observe that SFT on re-

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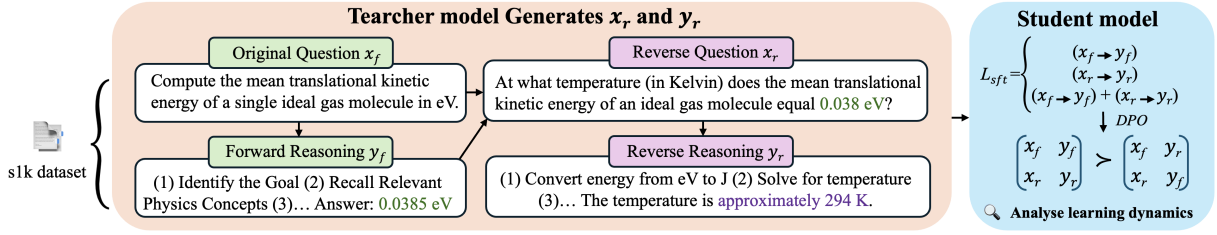


Figure 1: We begin with the s1k dataset (x_f, y_f) and generate reverse questions x_r , along with their corresponding reverse CoTs and answers y_r . We then fine-tune student models using cross-entropy loss under three settings as comparison: forward-only data, reverse-only data, and a mixture of both. To enhance directional consistency, we apply DPO to encourage directionally aligned responses while suppressing misaligned ones. Concurrently, we track the log probability of y_f and y_r across multiple fine-tuning stages to investigate the models’ learning dynamics.

verse data improves performance, whereas mixing forward and reverse examples leads to degradation. Mechanistic analysis shows that this reduces the model’s ability to distinguish reasoning paths. While Direct Preference Optimization (DPO; Rafailov et al. 2023) partly alleviates this, it still suffers from suboptimal initialization and tends to shift reverse reasoning probability toward irrelevant outputs. These findings underscore the need for improved alignment strategies to support robust reasoning.

2 Related work

Data-Efficient Reasoning in LLMs: s1 (Muenighoff et al., 2025), LIMO (Ye et al., 2025) and LIMA (Zhou et al., 2023) demonstrate that training on a small set of high-quality examples enables more effective performance, suggesting that massive datasets may not always be necessary to achieve competitive results. This perspective is further supported by methods such as iterative refinement (Madaan et al., 2023) and self-rewarding feedback (Asai et al., 2023), which demonstrate that reusing or distilling informative examples can improve model performance without relying on large-scale data. Complementary findings from data pruning and selection studies (Agarwal et al., 2024; Huang et al., 2024; Xu et al., 2025) reveal that indiscriminate scaling often yields diminishing returns, highlighting the value of targeted data curation in reasoning-intensive tasks.

Learning Dynamics of LLM Fine-Tuning: Neural Tangent Kernel (NTK) theory (Jacot et al., 2020; Arora et al., 2019) provides a framework for analyzing the influence of individual training examples during LLM fine-tuning. A gradient-based decomposition was later proposed (Ren and Sutherland, 2024), approximating the change in model confi-

dence for an output y on input x_o after training on a single example (x_u, y_u) as:

$$\Delta \log \pi_t(y | x_o) \approx -\eta A_t(x_o) K_t(x_o, x_u) G_t(x_u, y_u)$$

where K_t is the empirical NTK, G_t the gradient, and A_t a scaling factor tied to model certainty. This perspective helps explain interference, hallucination, and memorization (Pruthi et al., 2020). It also explains the diversity collapse (Dang et al., 2025), where correctness optimization concentrates the probability mass on a single reasoning path, limiting diversity. These insights inspire a learning-dynamics perspective on how mixed reasoning data shapes model behavior in multi-stage fine-tuning.

3 Methodology

3.1 Reverse Data Construction and Alignment

We begin with a forward reasoning dataset $\mathcal{D}_{s1k} = \{(x_f^{(i)}, y_f^{(i)})\}_{i=1}^{1000}$, consisting of 1,000 high-quality examples from the s1k dataset. Each example includes a question x_f and its corresponding CoT and answer y_f generated by Deepseek-R1 (R1). Based on each s1k’s question and final answer, we leverage R1 to construct the reverse dataset $\mathcal{D}_{r1k} = \{(x_r^{(i)}, y_r^{(i)})\}_{i=1}^{1000}$. For each forward example (x_f, y_f) , we prompt the model to generate a reverse question x_r that naturally elicits the original reasoning in reverse. Conditioned on x_r , we prompt R1 to generate the corresponding reverse reasoning chain and answer y_r . A concrete forward–reverse example is provided in Appendix A.5. We merge the above two datasets to obtain $\mathcal{D} = \mathcal{D}_{s1k} \cup \mathcal{D}_{r1k}$, which serves to investigate how bidirectional supervision influences model behavior and alignment.

We fine-tune the Qwen2.5-Instruct (Qwen et al., 2025) 7B and 14B models using the standard cross-entropy objective, where the input is the question x

and the target output y is the concatenation of the CoT and the final answer with special separation tokens.

Although SFT introduces forward and reverse reasoning, it does not equip LLMs with the ability to switch between two directions. To better align model responses with question directionality, we apply DPO following SFT. For this, we construct preference pairs of the form (x, y^+, y^-) , where x is the question, y^+ is the preferred response, and y^- is the response of reverse question. Specifically, for each example $(x_f, y_f) \in \mathcal{D}_{s1k}$, we treat the forward output as the preferred response, i.e., $y^+ = y_f$, and the corresponding reverse output as the rejected response, $y^- = y_r$. In contrast, for each example $(x_r, y_r) \in \mathcal{D}_{r1k}$, we assign $y^+ = y_r$ as the preferred response and $y^- = y_f$ as the rejected one. Each pair of preferences (x, y^+, y^-) is used to optimize the DPO objective, which encourages the model to prefer y^+ over y^- .

3.2 Analysis of the Pitfalls of Mixed Data

To investigate the fine-tuning behavior during both the SFT and DPO stages, we construct a small *probe training set* consisting of 100 examples: 50 forward instances $\mathcal{D}_f$ and their corresponding 50 reverse counterparts $\mathcal{D}_r$. The union of the two forms a mixed test dataset $\mathcal{D}_m = \mathcal{D}_f \cup \mathcal{D}_r$.

Throughout both the SFT and DPO stages, we monitor model behavior by recording intermediate checkpoints and evaluating the average log-probability (ALP) per token for both y^+ and y^- :

$$\begin{cases} \text{ALP}(y^+) = \frac{1}{N} \sum_{i=1}^N \frac{1}{|y_i^+|} \sum_{t=1}^{|y_i^+|} \log p(y_{i,t}^+ | x_i, y_{i,<t}^+), \\ \text{ALP}(y^-) = \frac{1}{N} \sum_{i=1}^N \frac{1}{|y_i^-|} \sum_{t=1}^{|y_i^-|} \log p(y_{i,t}^- | x_i, y_{i,<t}^-). \end{cases}$$

here, N denotes the number of examples in the *probe testing set*, and $|y_i^\pm|$ the length of each evaluated output, capped at 1000 tokens. This evaluation window is typically sufficient to capture the divergence between y^+ and y^- . Since the responses are long-form sequences, we normalize log-probability by sequence length to ensure fair comparison. Motivated by recent theoretical analyses of learning dynamics in LLMs (Ren and Sutherland, 2024), we track the margin:

$$\Delta = \text{ALP}(y^+) - \text{ALP}(y^-),$$

which serves as an empirical proxy for $\Delta \log \pi_t(y | x_o)$ in the NTK formulation, with ALP reflecting

model certainty $A_t(x_o)$, forward–reverse pairs indicating input similarity $K_t(x_o, x_u)$, and training supervision contributing gradient signals $G_t(x_u, y_u)$.

4 Experiments

We conducted experiments on the DeepSeek-R1’s s1k dataset, which consistently outperformed the Gemini-based variant. As the s1k has already been curated with quantity, diversity, and difficulty, we did not apply an additional filtering process. We fine-tune Qwen2.5-Instruct models (7B and 14B) in two stages on 8 A800-80GB GPUs. First, we applied SFT with LoRA (rank = 256, $\alpha = 512$). Then, we performed DPO using the trl library (von Werra et al., 2020), with DeepSpeed ZeRO-3 (Jacobs et al., 2023) and Flash Attention (Shah et al., 2024) to reduce memory usage. We employ open-source lm-eval-harness (Gao et al., 2024), with gpt-4o-mini to evaluate the accuracy.

4.1 Impact of Reverse and Mixed Data

To evaluate the effect of reverse data construction and bidirectional supervision, we conducted fine-tuning experiments on different training datasets. As shown in Table 1, we compare the distillation performance of models trained on the $\mathcal{D}_{s1k}$, $\mathcal{D}_{r1k}$, and $\mathcal{D}$ across three challenging benchmarks: AIME24-NoFigures (Mathematical Association of America, 2024), Math 500 (Lightman et al., 2023), and GPQA (Rein et al., 2024) benchmarks.

Table 1: Effect of Reverse Data ($\mathcal{D}_{r1k}$) and Mixed Training Sets ($\mathcal{D}$ and $\mathcal{D}_{0.5k}$) on downstream performance. Here, $\mathcal{D}_{0.5k}$ consists of 500 forward examples paired with their corresponding reverse data. In the same setting experiment, the best results are shown in bold.

Data	Model	AIME	Math	GPQA	Average
$\mathcal{D}_{s1k}$	3B	3.3%	45.2%	29.3%	25.9%
$\mathcal{D}_{r1k}$ (Ours)	3B	6.7%	43.4%	32.2%	27.5%
$\mathcal{D}_{0.5k}$	3B	0.0%	43.6%	29.8%	24.47%
$\mathcal{D}$	3B	0.0%	43.0%	26.2%	23.1%
$\mathcal{D}_{s1k}$	7B	16.7%	77.0%	34.0%	42.6%
$\mathcal{D}_{r1k}$ (Ours)	7B	20.0%	77.4%	42.4%	46.6%
$\mathcal{D}_{0.5k}$	7B	13.3%	71.8%	35.8%	40.3%
$\mathcal{D}$	7B	6.7%	56.0%	31.8%	31.5%
$\mathcal{D}_{s1k}$	14B	20.0%	83.2%	48.4%	50.6%
$\mathcal{D}_{r1k}$ (Ours)	14B	33.3%	86.0%	53.0%	57.4%
$\mathcal{D}$	14B	30.0%	81.6%	49.1%	53.6%

Under the same distillation pipeline, models trained on our reverse dataset $\mathcal{D}_{r1k}$ achieve an average improvement of 1.6%–6.8% compared to those trained on the original $\mathcal{D}_{s1k}$. However, combining

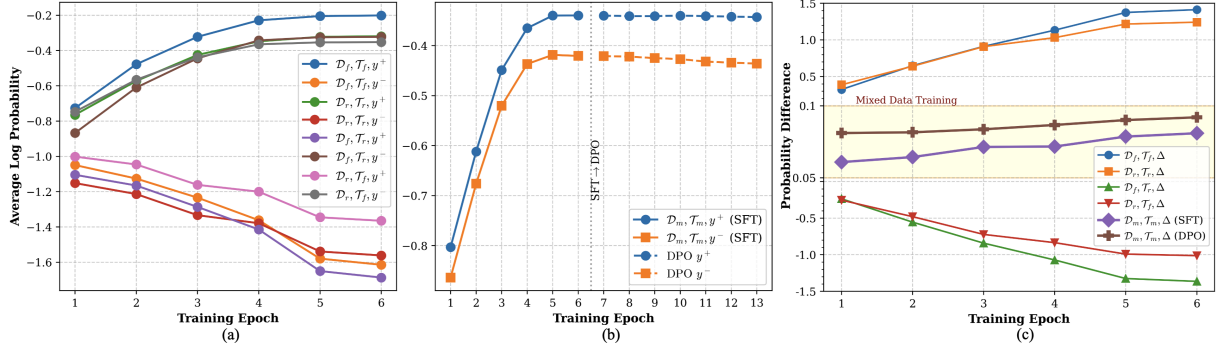


Figure 2: $\mathcal{D}$ denotes the training dataset, and $\mathcal{T}$ denotes the testing dataset. We report the Average Log Probability (ALP) for both the preferred responses (y^+) and the less preferred responses (y^-) in Figures (a) and (b), respectively. Figure (c) shows the difference $\text{ALP}(y^+) - \text{ALP}(y^-)$.

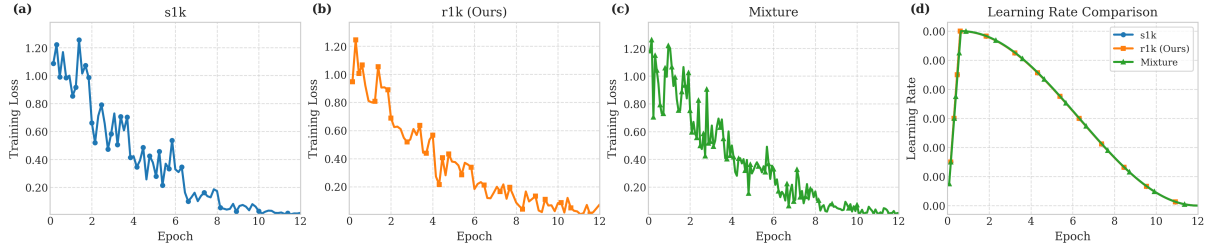


Figure 3: SFT Probe dataset Training Dynamics

forward and reverse examples leads to a significant drop in performance. As the size of the mixed dataset increases from $\mathcal{D}_{0.5k}$ to $\mathcal{D}$, the degradation becomes more pronounced, suggesting that mixed-direction reasoning data introduce interference between reasoning modes and hinder effective learning.

4.2 Teacher Model Reliability Analysis

We further compare SFT results when using Gemini 2.0 Flash Thinking (20–30B) versus DeepSeek-R1 (671B MoE) as the teacher model for generating chain-of-thought and answers. Table 2 shows that Gemini-based supervision leads to consistently 1–3% lower downstream accuracy compared to DeepSeek-R1 supervision. All experimental data are sourced from the $\mathcal{D}_{s1k}$ dataset.

Table 2: SFT based on two teacher models (Gemini and DeepSeek-R1). Parentheses indicate the teacher model used to generate the SFT reasoning and solution data.

Model / Teacher	AIME	GPQA	Math	Average
7B (Gemini)	13.3%	33.5%	77.4%	41.43%
7B (DeepSeek-R1)	16.7%	34.0%	77.0%	42.60%
14B (Gemini)	16.7%	47.2%	83.0%	48.93%
14B (DeepSeek-R1)	20.0%	48.4%	83.2%	50.60%

4.3 Impact of Directional Preference

Our DPO experiments use a temperature-weighting hyperparameter of $\beta = 0.6$, with the SFT-trained model fixed as the reference model. We apply DPO fine-tuning to four SFT-based models: three 7B models individually trained on $\mathcal{D}_{s1k}$, $\mathcal{D}_{r1k}$, and $\mathcal{D}$, and a 14B model trained on $\mathcal{D}$. All DPO models are further fine-tuned using preference pairs from $\mathcal{D}$, where each pair consists of two responses y^+ and y^- , generated from the opposite question.

Table 3: Effect of DPO on different SFT Data and based models settings. $\downarrow$ and $\uparrow$ indicate performance decrease and increase respectively; parentheses show relative change from the SFT baseline.

SFT Data	AIME	Math	GPQA	Average
$\mathcal{D}_{s1k}$ (7B)	13.3% $\downarrow$	71.8% $\downarrow$	35.9% $\downarrow$	40.3% ($\downarrow$ 2.3%)
$\mathcal{D}_{r1k}$ (7B)	16.7% $\downarrow$	75.4% $\downarrow$	39.4% $\downarrow$	43.8% ($\downarrow$ 2.8%)
$\mathcal{D}$ (7B)	16.7% $\uparrow$	64.2% $\uparrow$	34.8% $\uparrow$	38.6% ($\uparrow$ 7.1%)
$\mathcal{D}$ (14B)	40.0% $\uparrow$	81.2% $\downarrow$	46.4% $\downarrow$	55.9% ($\uparrow$ 2.3%)

Table 3 shows that applying DPO with mixed preference data on the $\mathcal{D}_{s1k}$ (7B) and $\mathcal{D}_{r1k}$ (7B) reference models leads to a performance decline. For the model initially fine-tuned on the mixed dataset $\mathcal{D}$, DPO achieves some performance improvements, but its overall performance remains inferior to SFT trained on $\mathcal{D}_{r1k}$.

4.4 Analysis of the Pitfalls of Mixed Data

We analyze the in-distribution pairs $(\mathcal{D}_f, \mathcal{T}_f)$, $(\mathcal{D}_r, \mathcal{T}_r)$, and $(\mathcal{D}_m, \mathcal{T}_m)$, where $\mathcal{D}$ and $\mathcal{T}$ denote the training and testing datasets respectively. SFT is run for 12 epochs with evaluation every 2 epochs, and DPO for 7 epochs with evaluation after each. Changes in the ALP of y^+ reflect the learned strategy, while y^- indicates hallucination. We also consider the out-of-distribution pairs $(\mathcal{D}_f, \mathcal{T}_r)$ and $(\mathcal{D}_r, \mathcal{T}_f)$, where variations in the ALP of y^+ measure the generalization capability to handle reverse question, whereas y^- indicate the likelihood of generating irrelevant or off-target responses.

Figure 2(a) shows that under out-of-distribution scenarios, models trained on the $\mathcal{D}_r$ exhibit lower hallucination rates y^- and better generalization. For in distribution settings, (a) demonstrates that the likelihood of y^+ increase significantly, but this improvement is accompanied by a corresponding rise in hallucinations y^- . Figure 2(b) reveals that mixed-data training $\mathcal{D}_m$ induces a stronger increase in hallucinations, while the likelihood of preferred responses fails to reach the levels achieved by training solely on $\mathcal{D}_r$ and $\mathcal{D}_f$. Even though the subsequent DPO improves preference alignment by suppressing the probability of y^- to irrelevant responses, this suppression is limited.

Figure 2(c) shows that models trained on $\mathcal{D}_f$ and $\mathcal{D}_r$ maintain a gap between y^+ and y^- , whereas the model trained on mixed data $\mathcal{D}_m$ produces only a narrow margin (0.05–0.1). This suggests that SFT on mixed data weakens LLM’s ability to discriminate the learned strategies and hallucination. Although DPO slightly separates y^+ and y^- , the effect remains limited. We hypothesize that the conflicting signals from $\mathcal{D}_m$ lead the model to optimize in competing directions, hindering the formation of coherent preferences. This phenomenon also helps explain why models trained on smaller but higher-quality datasets, such as LIMO or s1k, can outperform larger ones: consistent supervision leads to more effective optimization.

5 Conclusion

We constructed a high-quality reverse reasoning dataset r1k and demonstrated its effectiveness in improving reasoning ability. We further investigate the effects of mixed data during multi-stage fine-tuning, underscoring the need for improved alignment strategies to support robust reasoning.

6 Limitations

This work explores the integration of reverse reasoning data in multi-stage fine-tuning, but several limitations remain. Our reverse dataset $\mathcal{D}_{r1k}$ is constructed by automated prompting without human validation, which may introduce subtle errors or inconsistencies in reasoning quality. Additionally, the DPO formulation assumes a strict directional preference between forward and reverse outputs, potentially oversimplifying cases where both reasoning directions offer complementary insights. Furthermore, while this study adopts a standard SFT + DPO pipeline, alternative alignment strategies may offer more robust solutions to conflicting supervision in mixed data settings.

7 Ethics Statement

This work trains and evaluates models on publicly available datasets (s1k, GPQA, AIME, and Math 500), each used in accordance with its respective license. We release our $\mathcal{D}_{r1k}$ dataset and code strictly for research purposes. In this study, no personal or sensitive information is collected or used.

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A Appendix

A.1 Reverse Data Construction

We design two prompt templates to support our bidirectional reasoning framework. These templates serve complementary purposes in data construction, enabling the model to learn both forward and reverse reasoning patterns. The first template is designed to generate reverse question-answer (QA) pairs. Given a question and its corresponding answer, the model is instructed to invert their roles: reformulating the answer into a thought-provoking question and the original question into a reflective statement that serves as the new answer.

```
"role_definition":
"You are an AI model tasked with generating a
reflective thinking exercise.
Given the following question and answer:"
```

```
- Question: {question}
- Answer: {answer}
```

```
"instructions":
"Your task is to reverse the roles of the
question and answer.
Transform the answer into a question that
is thought-provoking and encourages deeper
reflection.
Similarly, convert the original question into
a statement that serves as an insightful
answer.
Ensure that the new question remains
reasonable and stimulates further inquiry,
while the new answer is right to the question."
```

```
"expected_output":
- New Question:
- New Answer:
```

The second template focuses on generating step-by-step solutions with CoT prompting, followed by a clearly demarcated final answer. This template is specifically designed to extract DeepSeek R1’s detailed reasoning chains and final answers, serving as the foundation for producing high-quality reasoning examples.

```
"role_definition":
"You are an AI model that is designed to
generate solutions to a given question.
All numerical answers must be explicitly
marked with \boxed{}."
```

```
- Question: {question}
```

```
"instructions":
"Ensure your answer is absolutely correct and
standard."
```

```
"expected_output":
Presents the complete and concise answer.
If the answer contains only one numerical
value, it must be marked in the form of
\boxed{}."
```

The r1k dataset shares the same domains as the s1k dataset (Muennighoff et al., 2025), and its token length distribution is shown in Figure 4. We select the 50 pieces of data with the least text tokens in s1k and the corresponding r1k data as our probe dataset. All external datasets used in our work are publicly available and licensed for research use. Specifically, AIME and OpenAI’s Math 500 datasets are distributed under the Apache License 2.0, while GPQA is released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.

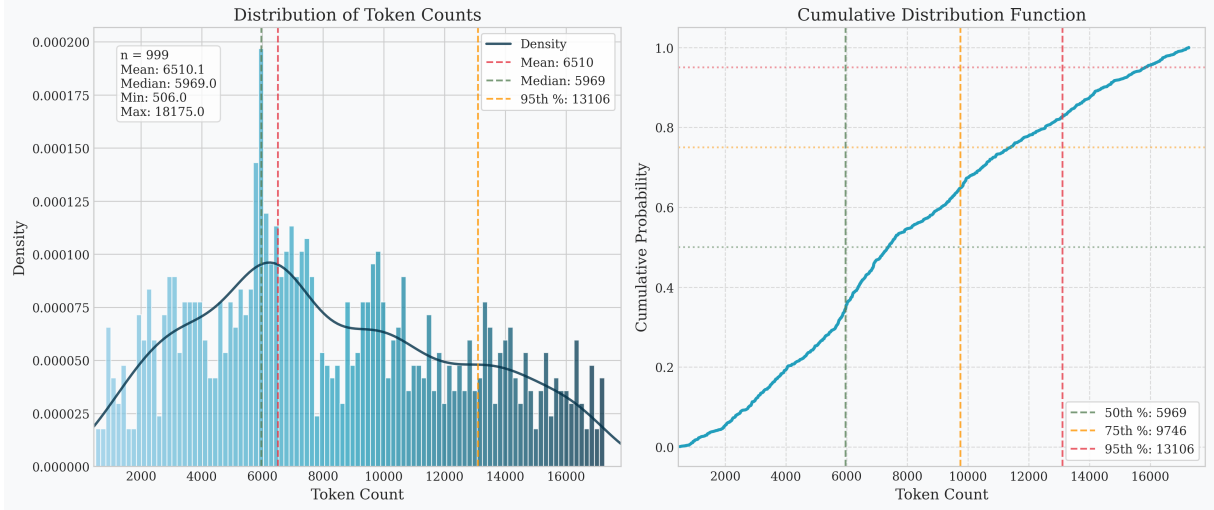


Figure 4: Distribution of r1k Token Counts

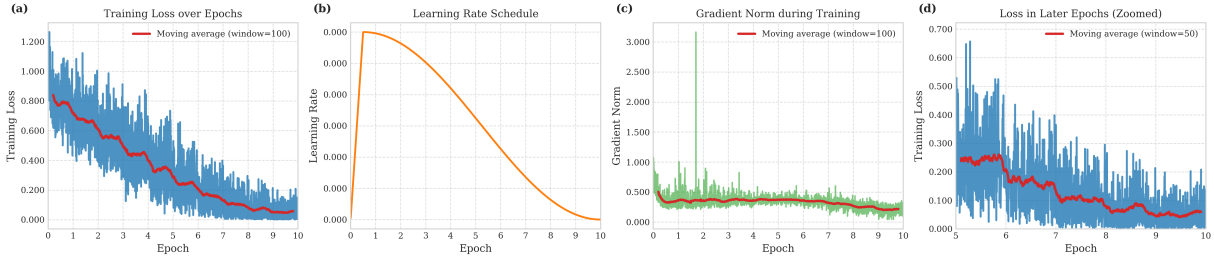


Figure 5: SFT Mixed data Training Details

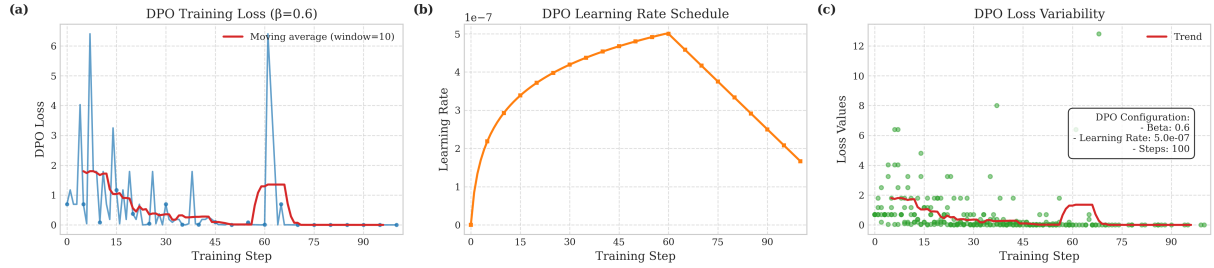


Figure 6: DPO-Training Details (First 100 steps)

A.2 Training Details

The training curves in Figures 5–6 illustrate learning dynamics under different supervision settings by tracking key optimization parameters of loss, learning rate, and gradient norm. Figure 5 depicts the progression of SFT on mixed data, while Figure 6 provides a probe-based analysis across forward, reverse, and mixed reasoning styles. Figure 6 captures the early-stage DPO optimization, emphasizing how preference alignment evolves within the first 100 training steps.

A.3 Additional Experimental Results

A.3.1 DPO with Separate SFT Stage

In addition to the SFT and DPO pipeline, we also evaluate applying DPO directly on the mixed dataset $\mathcal{D}$ (2k examples from s1k and r1k), without a prior SFT stage. Results in Table 4 show that direct DPO yields lower performance than SFT followed by DPO, confirming that SFT provides a stable initialization.

A.4 Experiment Settings

We adopt a two-stage training pipeline consisting of supervised SFT followed by DPO. This section details our model configurations, optimization strategies, and resource settings.

Table 4: Direct DPO (without SFT) on mixed datasets $\mathcal{D}$ (2k data from slk and r1k).

Model	AIME	GPQA	Math	Average
7B	10.0%	34.6%	59.0%	34.53%
14B	20.0%	40.2%	72.0%	44.07%

A.4.1 Supervised Fine-Tuning (SFT)

We use the Qwen2.5-7B-Instruct model and train on 4 A800 GPUs, with a maximum sequence length of 20,000 tokens. Some of the excess data are replaced with Gemini data for compatibility. For the Qwen2.5-14B-Instruct model, we performed training on 8 A800 GPUs under the same sequence-length setting. Training 7B-based models with 1k data takes around 3 hours, while 14B-based models requires 6-7 hours.

We adopt a cosine learning rate schedule with restarts and set the initial learning rate to 3×10^{-4} . Other hyperparameters include:

- Weight decay: 1×10^{-6}
- Adam optimizer: $\beta_1 = 0.9, \beta_2 = 0.95$
- Gradient clipping: 1.0
- Warmup ratio: 0.05

Training Configuration.

- Number of epochs: 10
- Per-device batch size: 1
- Gradient accumulation steps: 1
- Mixed precision: BF16

Parameter-Efficient Fine-Tuning. We apply LoRA with the following configuration:

- Rank $r = 256$, scaling factor $\alpha = 512$
- Target modules: "q_proj, k_proj, v_proj, o_proj, gate_proj, down_proj, up_proj, lm_head"
- Optimizer: 8-bit AdamW

A.4.2 Direct Preference Optimization (DPO)

We initialize from the SFT checkpoint and train on 4 GPUs. The training runs for a maximum of 200 steps in around 3 hours. We experiment with DPO using the value of β ranging from 0.1 to 0.8, and observe that $\beta=0.6$ leads to the most significant performance improvement on mixed-direction data.

However, under this setting, the change in average log-probability (ALP) is relatively subtle. To better highlight learning dynamics in our probe-based analysis, we use a smaller $\beta=0.2$, which yields more distinguishable ALP shifts between y^+ and y^- .

Optimization Settings.

- Learning rate: 5×10^{-7} with cosine decay
- KL penalty coefficient $\beta = 0.6$ for DPO training, $\beta = 0.2$ for probe dataset training and testing.

Training Configuration.

- Per-device batch size: 1
- Gradient accumulation: 4 steps
- Mixed precision: BF16
- Maximum prompt length: 600 tokens
- Maximum sequence length: 20,000 tokens

Efficiency Enhancements. We enable the following optimizations to reduce memory and computation overhead:

- Flash Attention
- Gradient checkpointing
- DeepSpeed ZeRO-3 with parameter sharding

Distributed Optimization. We apply DeepSpeed ZeRO-3 for memory efficiency and optimize multi-GPU communication through NCCL tuning.

A.5 Example of Forward–Reverse Data Construction

As an illustrative case, we present a representative forward–reverse pair from $\mathcal{D}_{slk}$ and $\mathcal{D}_{r1k}$. The forward instance is taken directly from $\mathcal{D}_{slk}$, whereas the reverse instance is obtained by inverting the question–answer direction and generating the corresponding chain-of-thought with DeepSeek-R1.

A.5.1 Forward Example (from $\mathcal{D}_{slk}$).

Question: Find any quadruple of positive integers (a, b, c, d) satisfying $a^3 + b^4 + c^5 = d^{11}$ and $abc < 10^5$.

Answer: It’s easy to guess that there are solutions such that a, b, c, d are in the form of n^x , where n is a rather small number. After a few attempts,

we can see that we obtain simple equations when $n = 2$ or $n = 3$: for $n = 2$, the equation becomes in the form of $2^t + 2^t + 2^{t+1} = 2^{t+2}$ for some non-negative integer t ; for $n = 3$, the equation becomes in the form of $3^t + 3^t + 3^t = 3^{t+1}$ for some non-negative integer t . In the first case, we hope that t is a multiple of two of 3, 4, 5, that $t + 1$ is a multiple of the last one, and that $t + 2$ is a multiple of 11 . Therefore, $t \equiv 15, 20, 24 \pmod{60}$ and $t \equiv 9 \pmod{11}$. It's easy to check that the only solution that satisfies the given inequality is the solution with $t = 20$, and $(a, b, c, d) = (128, 32, 16, 4)$. In the case where $n = 3$, we must have that t is a multiple of 60 , which obviously doesn't satisfy the inequality restriction. Remark: By programming, we find that the only two solutions are $(a, b, c, d) = (128, 32, 16, 4)$ and $(a, b, c, d) = (160, 16, 8, 4)$, with the the former being the intended solution.

A.5.2 Reverse Example (from $\mathcal{D}_{r1k}$).

Question: The quadruples $(128, 32, 16, 4)$ and $(160, 16, 8, 4)$ satisfy a specific equation involving exponents of the form $a^k + b^m + c^n = d^p$, along with the condition $abc < 10^5$. What is the exact equation (including the exponents k, m, n, p) and the constraint?

Answer: The equation satisfied by both quadruples is $a^3 + b^4 + c^5 = 4^{11}$ with the constraint $abc < 10^5$. For the first quadruple $(128, 32, 16, 4)$:
- $128^3 + 32^4 + 16^5 = 2^{21} + 2^{20} + 2^{20} = 2^{22} = 4^{11}$.
- $abc = 128 \times 32 \times 16 = 65,536 < 10^5$.
For the second quadruple $(160, 16, 8, 4)$:
- $160^3 + 16^4 + 8^5 = (16 \times 10)^3 + 2^{16} + 2^{15} = 4,096,000 + 65,536 + 32,768 = 4,194,304 = 4^{11}$.
- $abc = 160 \times 16 \times 8 = 20,480 < 10^5$. Thus, the exact equation is $a^3 + b^4 + c^5 = 4^{11}$ with the constraint $abc < 10^5$.

Chain-of-thought (concise overview):

Observe $128 = 2^7$, $32 = 2^5$, $16 = 2^4$, $4 = 2^2$. Enforce $5m = 4n$ so the latter two terms match in exponent; taking $m = 4$, $n = 5$ gives both as 2^{20} . Choose $k = 3$ so $2^{7k} = 2^{21}$, and the sum becomes $2^{21} + 2 \cdot 2^{20} = 2^{22} = 4^{11}$, hence $p = 11$. The same exponents verify the second quadruple.